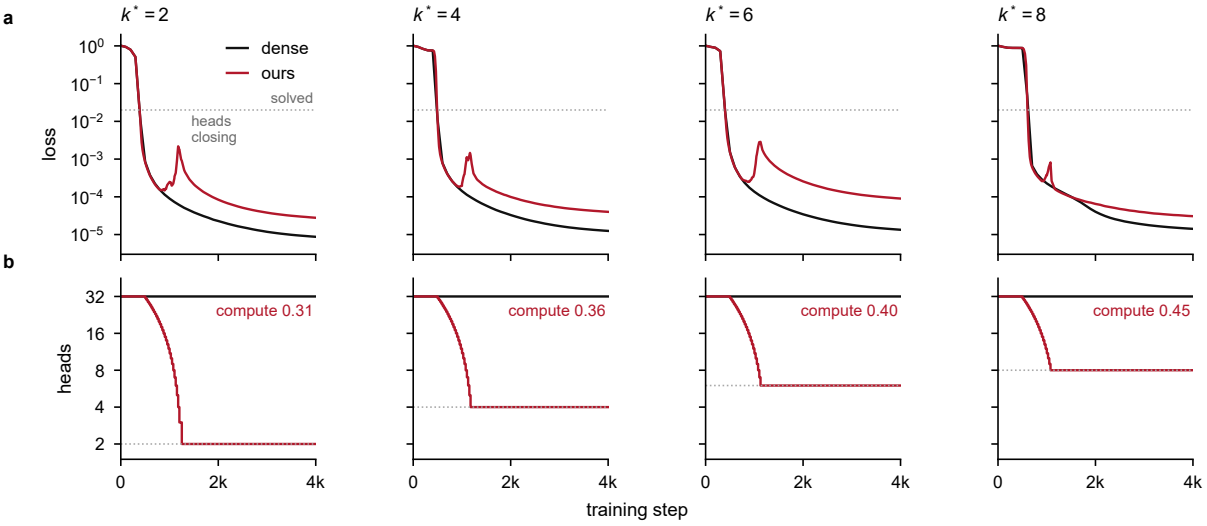


c

Method	Head score	During training	Sets own size	Reopens	exact count / never broke
Hemodynamic attenuation (ours)	$\min_W L_{-h} - \min_W L$ Optimal Brain Surgeon (Hassibi & Stork 1993)	●	●	●	29/30 30/30
Published methods					
Kurtic et al. 2023 (ZipLM)	$\min_W L_{-h} - \min_W L$ Optimal Brain Surgeon (Hassibi & Stork 1993)	–	–	–	24/30 0/30
Michel et al. 2019	$\lvert \partial L / \partial g_h \rvert$ gradient of the head gate	–	–	–	28/30 0/30
Voita et al. 2019	$L + \lambda \Sigma_h P(g_h \neq 0)$ a learned gate per head (L0 penalty)	–	–	–	9/30 0/30
Controls: our method with one part removed					
no fade (instant closing)	$\min_W L_{-h} - \min_W L$ Optimal Brain Surgeon (Hassibi & Stork 1993)	●	●	●	30/30 0/30
no re-fit	$L_{-h}(W) - L(W)$ Optimal Brain Damage (LeCun et al. 1989)	●	●	●	2/30 30/30
no score (random choice)	random head	●	●	●	30/30 6/30

Tasks needing 2, 4 and 8 heads, 10 seeds each. ● yes – no

